Fair for Whom?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Fair for Whom? before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Missing and Unbalanced Examples

Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Recommendations

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prediction Is Not Fact

Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

Complete the supported Fair for Whom? trace and record the first step you can explain without help.

Use W2: Compare totals and conditions for Fractions and performance comparisons. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Fair for Whom? explanation against the supplied facts before discussing it.

Use the worked case for Fractions and performance comparisons. Proposed explanation: “Only correct cases belong in the **denominator**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Compare a new shared sound test

Versions X and Y label the same 20 previously unseen clips: 12 quiet and 8 noisy, with no missing outcomes. X is correct on 10 quiet and 2 noisy clips; Y on 9 quiet and 5 noisy clips.

Calculate each combined fraction and percentage, identify the overall leader and compare the leaders within each group. Explain why an overall result cannot guarantee the next clip's outcome.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Fair for Whom? target to a trace and a justified conclusion.

System A succeeds on 6 of 8 quiet tests and 3 of 4 noisy tests. Is its noisy success rate lower? Show fractions and percentages. What can these small tests not establish?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Fair for Whom? case and show the deciding evidence.

A voice system succeeds on 8/10 quiet and 4/10 noisy recordings.

Compare and propose the next investigation.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A helper succeeds on 4 of 4 clear voices and 1 of 4 quiet voices. Who is less well served? What **evidence** should a repair preserve?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 23 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Fair for Whom? adds to Helpful Classroom Sorter and keep the evidence.

Test the sorter across relevant item conditions and users; never infer sensitive identity to create a **fairness** check.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

> System A succeeds on 6 of 8 quiet tests and 3 of 4 noisy tests. Is its noisy success rate lower? Show fractions and percentages. What can these small tests not establish?

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain both parts of a performance fraction

> Explain both parts of a performance fraction. Apply the supplied Fair for Whom? facts and keep your reasoning.

For version A in the lesson, write correct and assessed counts, simplify the **accuracy** fraction and express it as a percentage. Explain what its **denominator** includes and why 9/9 would be misleading.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Compare totals and conditions

> Compare totals and conditions. Apply the supplied Fair for Whom? facts and keep your reasoning.

Compare A and B on the shared 12-card test and on its bright and dim groups. State the leader in each and explain why the overall result does not imply that one version wins in every condition.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Do not hide unassessed outcomes

> Do not hide unassessed outcomes. Apply the supplied Fair for Whom? facts and keep your reasoning.

C has 9 correct and 1 incorrect assessed outcome, with 2 other selected cases unassessed. Write its assessed **accuracy** and missing count. A reviewer reports 9/12 as assessed **accuracy**; another reports 90% on all 12.

Explain the problem with each label.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Compare a new shared sound test

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Compare a new shared sound test. Apply the supplied Fair for Whom? facts and keep your reasoning.

Versions X and Y label the same 20 previously unseen clips: 12 quiet and 8 noisy, with no missing outcomes. X is correct on 10 quiet and 2 noisy clips; Y on 9 quiet and 5 noisy clips.

Calculate each combined fraction and percentage, identify the overall leader and compare the leaders within each group. Explain why an overall result cannot guarantee the next clip's outcome.

My reasoning and evidence

W5 Combine counts rather than group percentages

> Combine counts rather than group percentages. Apply the supplied Fair for Whom? facts and keep your reasoning.

A separate test has 2 correct out of 2 assessed in group P and 4 correct out of 8 assessed in group Q. Calculate combined **accuracy** as a fraction and percentage. A chart reports 75% by averaging the group percentages.

Explain why that is not the **accuracy** over all assessed cases.

> **My reasoning and evidence**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Fair for Whom?

> A fictional badge reader is evaluated on bright and dim cards with known reference labels. Explain the labelled step before checking the changed condition.

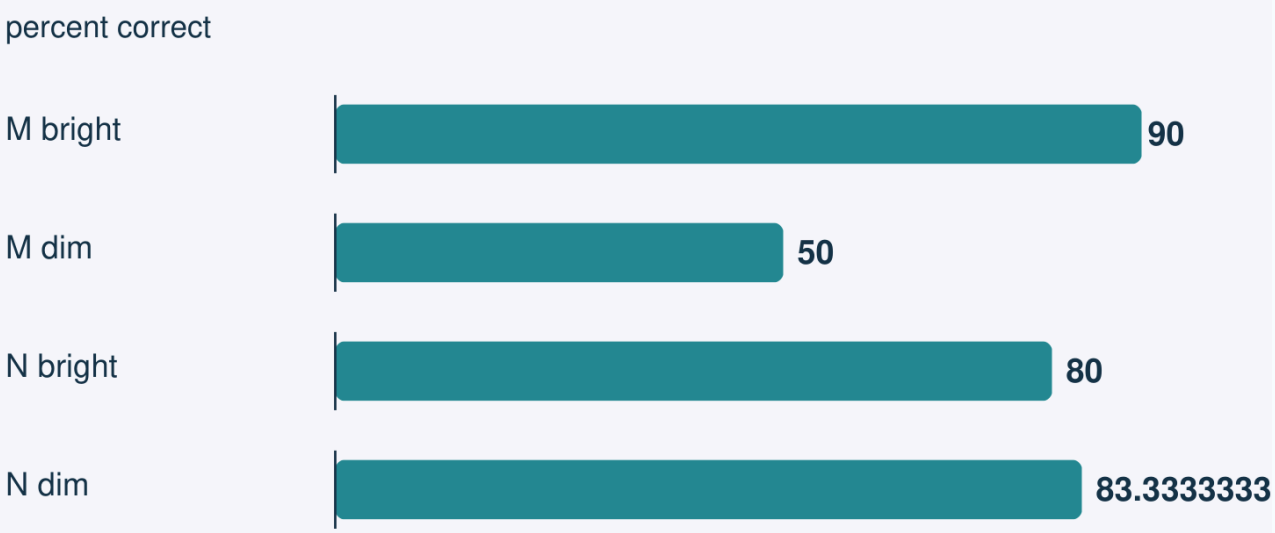
A fictional badge reader is evaluated on bright and dim cards with known reference labels.

Version M is correct on 9 of 10 bright cards and 3 of 6 dim cards.

Version N uses the same cards: 8 of 10 bright and 5 of 6 dim correct.

AI ACADEMY / AI-1/23

Group rates compared on the same cards



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/23

Fair for Whom?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Bright group

M:9/10=90%; N:8/10=80%.



2. Dim group

M:3/6=50%; N:5/6≈83.3%.



3. Combined counts

M:(9+3)/16=75%; N:(8+5)/16=81.25%.



4. Discuss trade-off

N improves total and dim results but loses one bright success.

CHANGE ONE THING

Give bright and dim groups equal weight as an explicitly different metric.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which group loses performance under N, and can total improvement erase that fact?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional badge reader is evaluated on bright and dim cards with known reference labels.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? N improves total and dim results but loses one bright success.

1. Dim group

2. Combined counts

3. Bright group

4. Discuss trade-off

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Discuss trade-off N improves total and dim results but loses one bright success. Adding correct counts and **denominator**s weights each tested card once.

Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

Decide what matters with the relevant users rather than announcing that one number proves **fairness**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Give bright and dim groups equal weight as an explicitly different metric. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. M scores **70%**; N scores about 81.7%.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: M scores **70%**; N scores about

81.7%. This averages group rates and is not the original 16-card agreement rate.

Repair a missing building block

Review Missing and Unbalanced Examples (ai-1/22). Goal: Name the target collection, selection list and cases actually assessed. Short practice: Would testing only dim plain cards close every current gap?

Point to the target groups.. Return to this week and explain what you changed.

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone's preferences?.

Return to this week and explain what you changed.

Review **Prediction** Is Not Fact (ai-1/17). Goal: Separate a recorded **prediction** from the **observation** used to check it. Short practice: Which field changes the **observation** label, and which field changes the model **prediction**?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

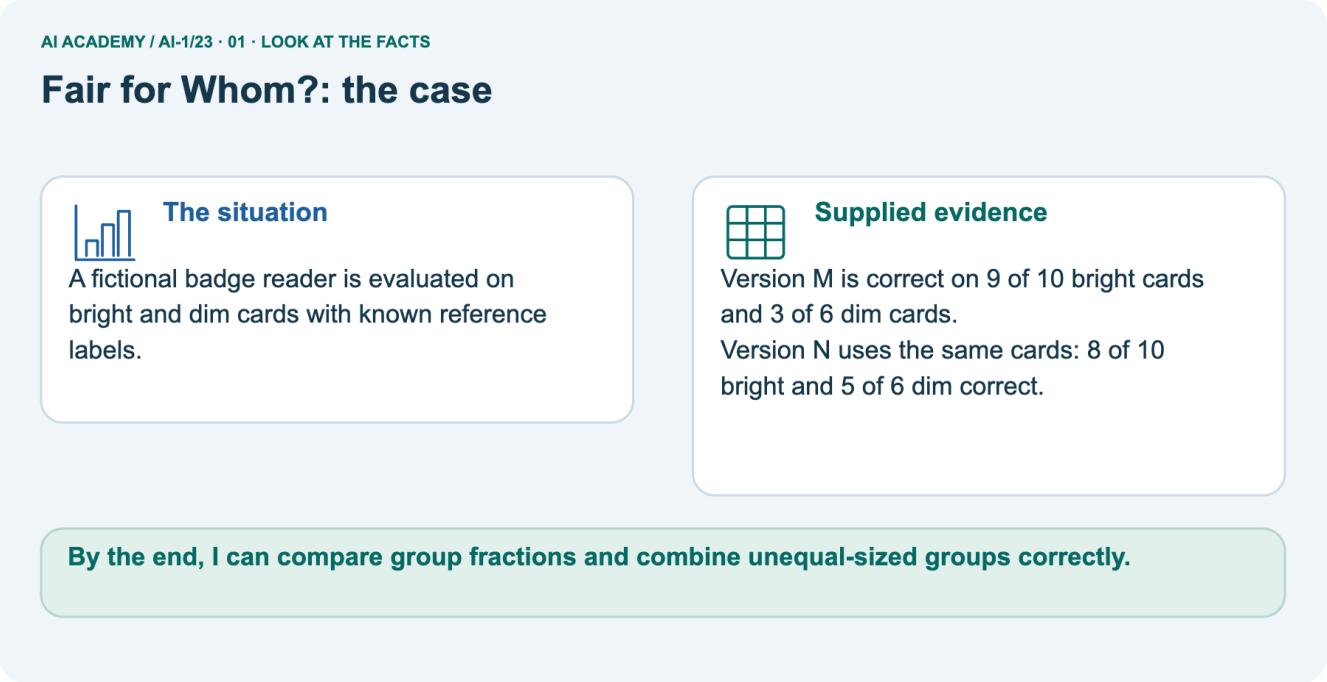
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Fair for Whom? picture from its labels, then record a relevant limit.

01. Fair for Whom?: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

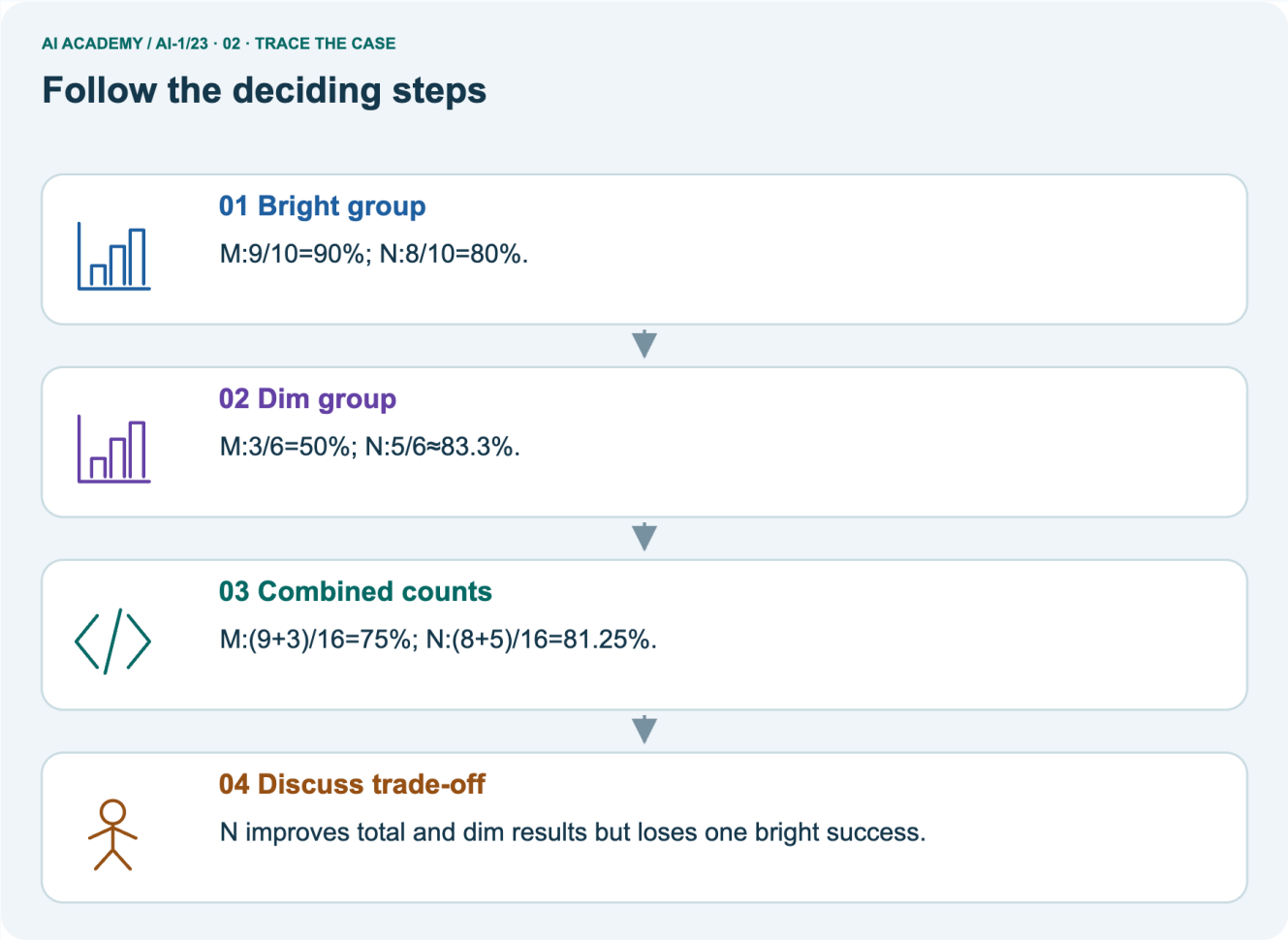
A fictional badge reader is evaluated on bright and dim cards with known reference labels. Supplied facts: Version M is correct on 9 of 10 bright cards and 3 of 6 dim cards.

Version N uses the same cards: 8 of 10 bright and 5 of 6 dim correct.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



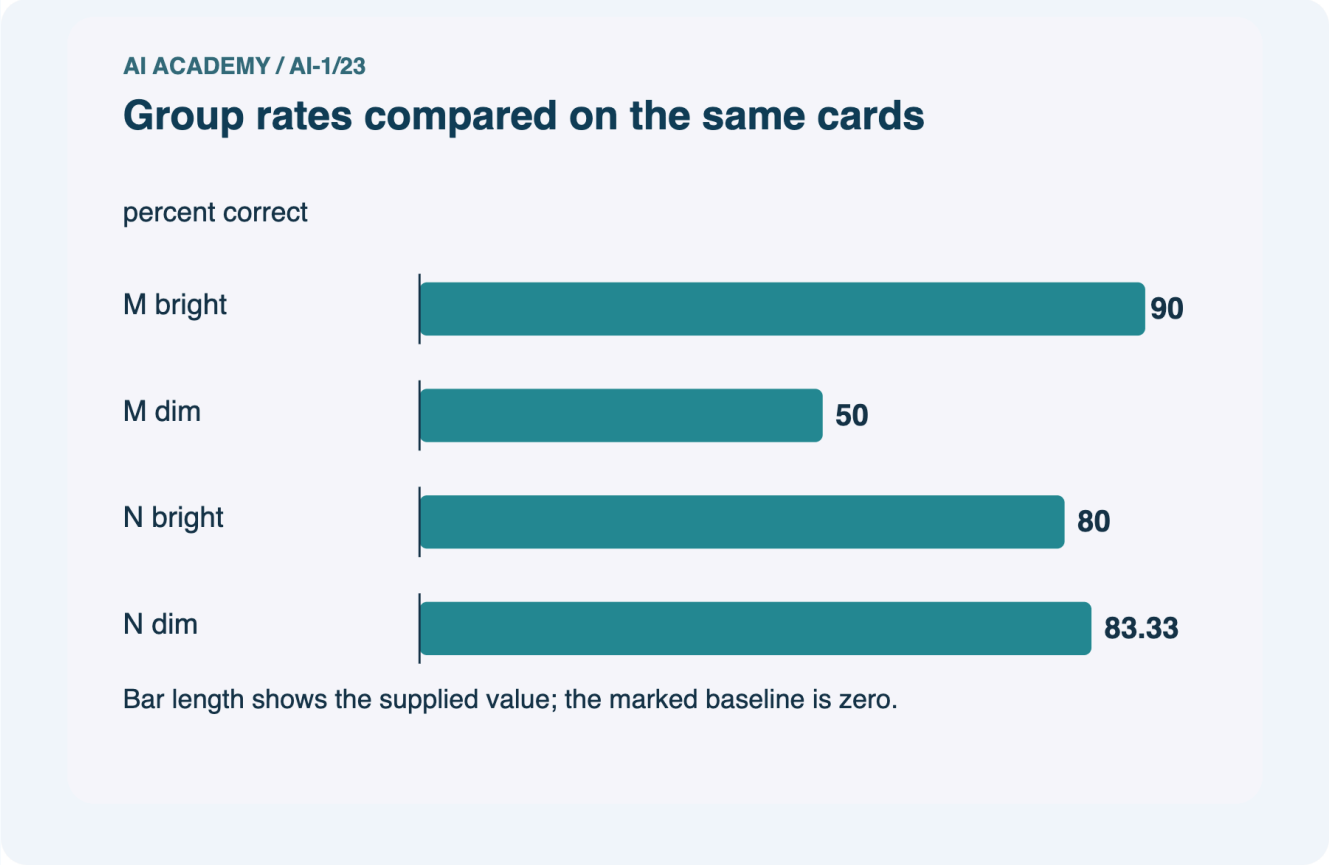
Follow the labelled picture with the matching explanation. Open the vector diagram

Bright group: M:9/10=90%; N:8/10=**80%**. → Dim group: M:3/6=50%; N:5/6≈83.3%. → Combined counts: M:(9+3)/16=75%; N:(8+5)/16=81.25%. → Discuss trade-off: N improves total and dim results but loses one bright success.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Group rates compared on the same cards



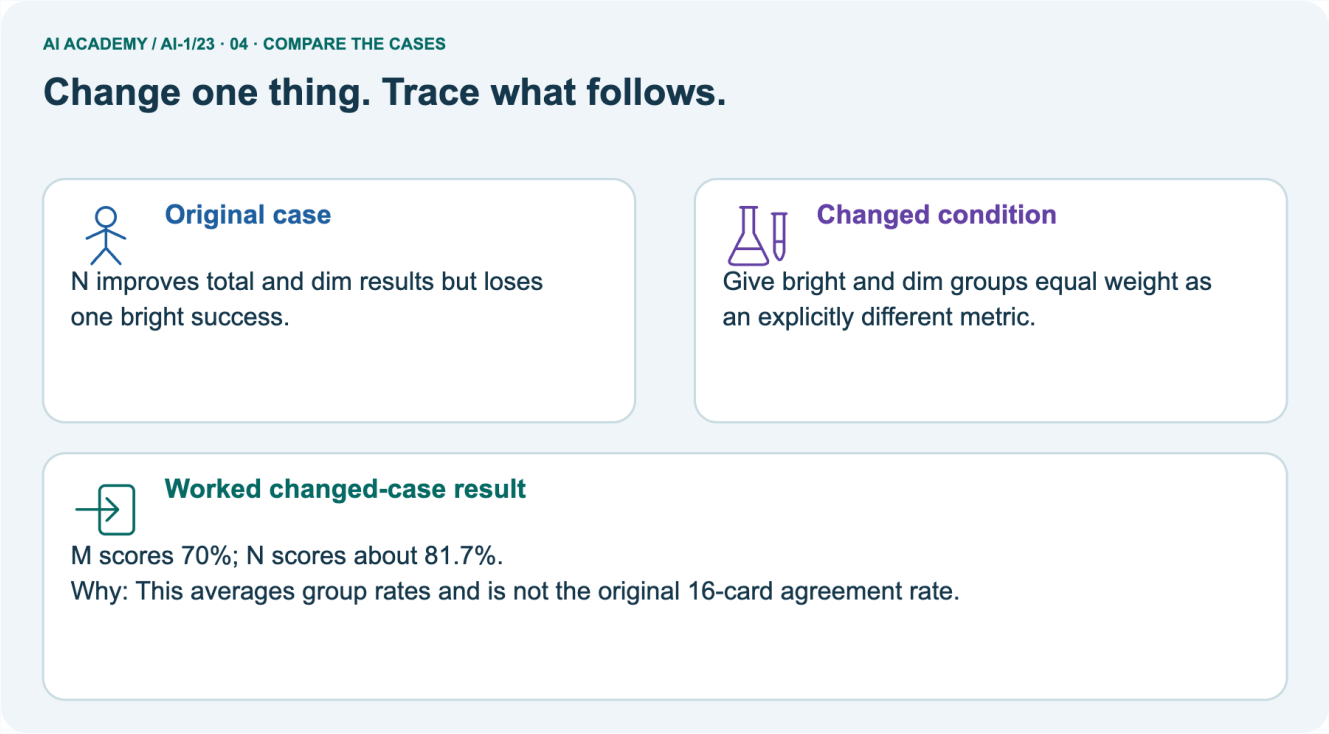
Follow the labelled picture with the matching explanation. Open the vector diagram

Group rates compared on the same cards. Units: percent correct. M bright: 90. M dim: 50. N bright: 80. N dim: 83.33333333333333.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

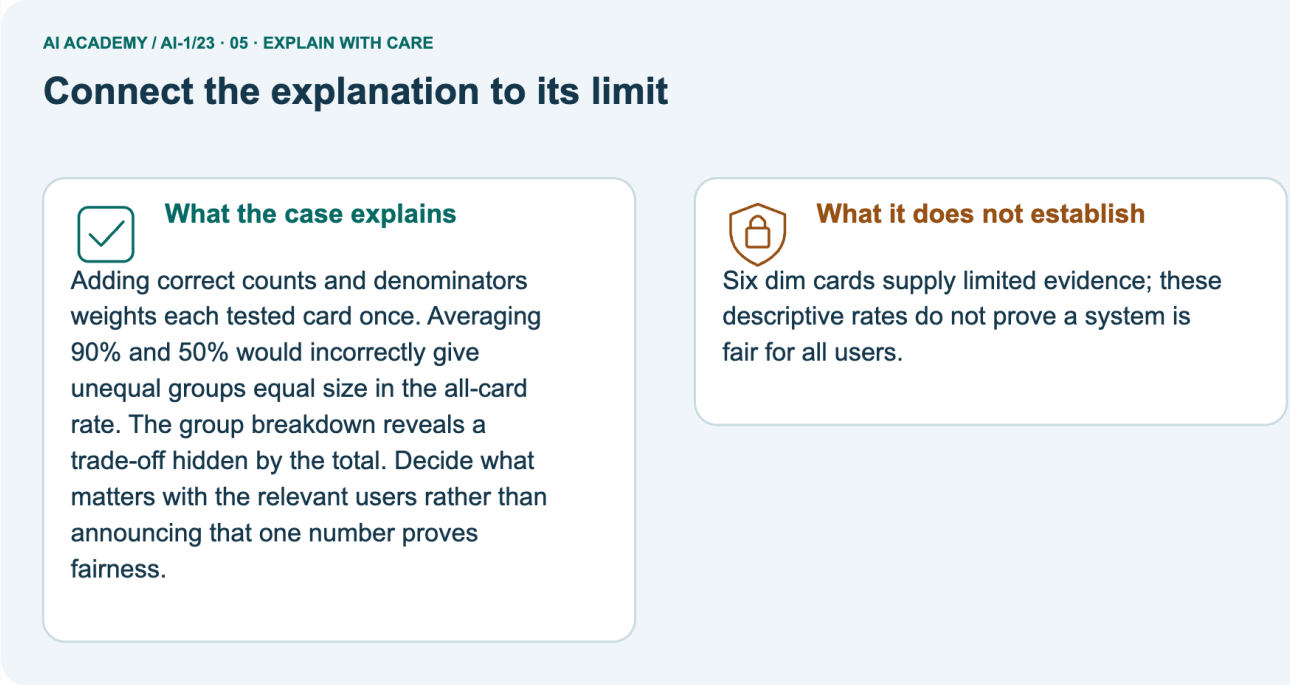
Original case: N improves total and dim results but loses one bright success. Changed condition: Give bright and dim groups equal weight as an explicitly different metric. Expected result: M scores **70%**; N scores about 81.7%.

Reason: This averages group rates and is not the original 16-card agreement rate.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

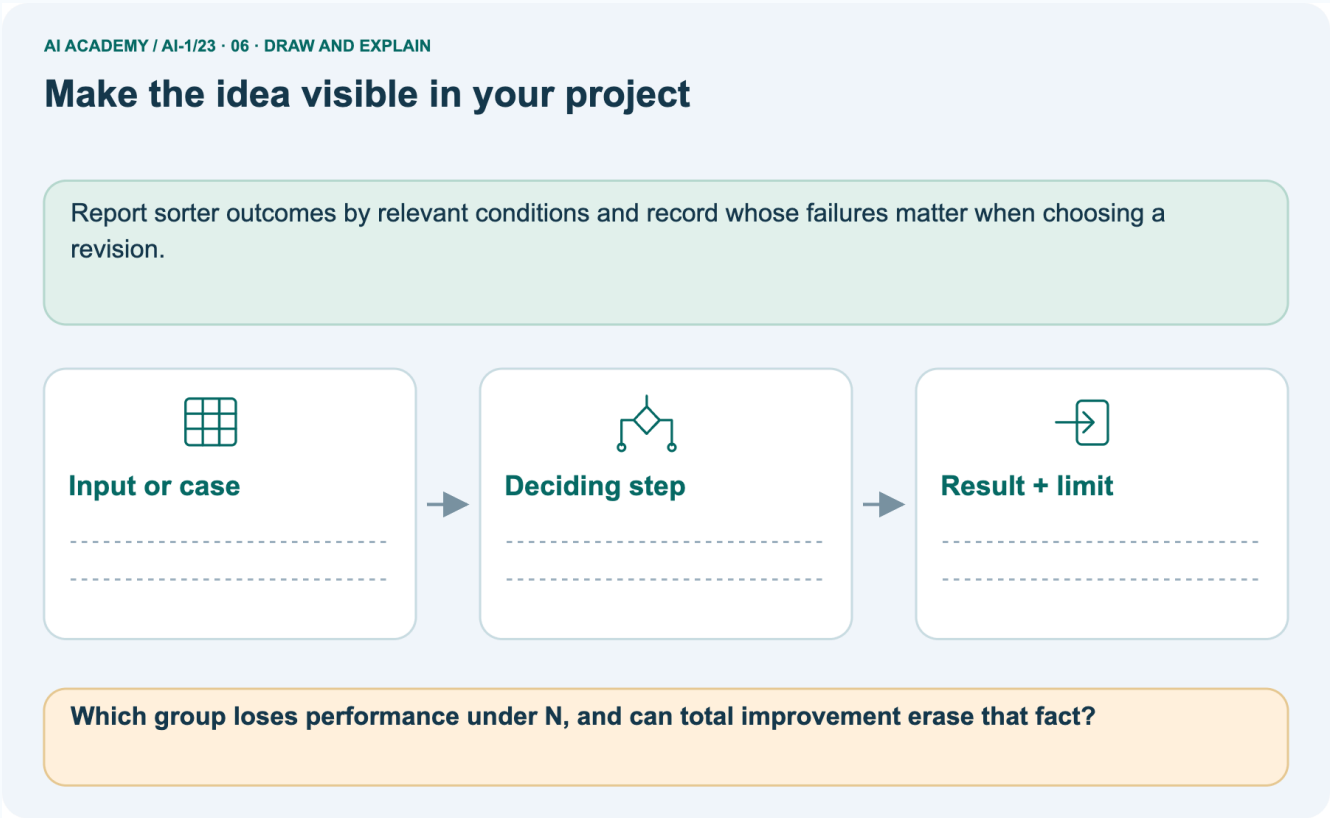
Adding correct counts and **denominator**s weights each tested card once. Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

Decide what matters with the relevant users rather than announcing that one number proves **fairness**. Limit: Six dim cards supply limited **evidence**; these descriptive rates do not prove a system is fair for all users.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision. Draw your **input**, deciding step, result and limit.

Supported question: Which group loses performance under N, and can total improvement erase that fact?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.